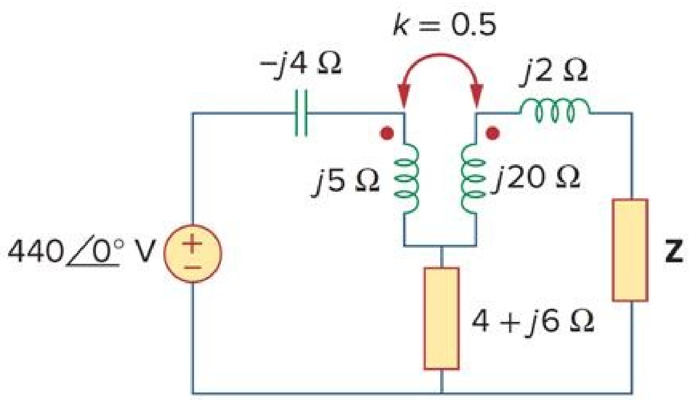
Chapter 13, Problem 18P

Problem

Find the Thevenin equivalent to the left of the load Z in the circuit of Fig. 13.87.

Figure 13.87



Step-by-step solution

Step 1 of 8

Refer to Figure 13.87 for the circuit in the text book.

Redraw the circuit in the phasor domain as shown in Figure 1. Determine the Thevenin equivalent circuit across the terminals *a* and *b*.

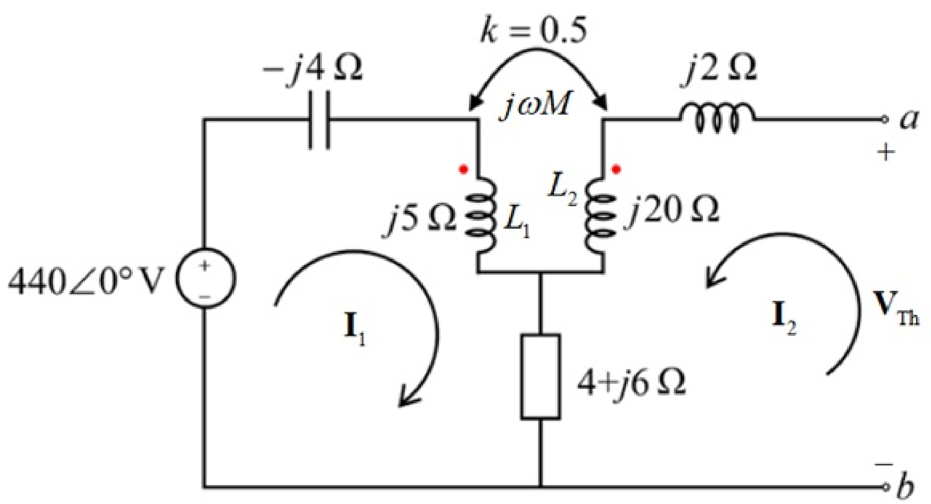


Figure 1

Step 2 of 8

The impedance of inductor L_1 is,

$$j\omega L_1 = j5\ \Omega$$
$$\omega L_1 = 5$$

The impedance of inductor L_2 is,

$$j\omega L_2 = j20\ \Omega$$
$$\omega L_2 = 20$$

Determine the impedance of the mutual inductance.

$$k = \frac{M}{\sqrt{L_1 L_2}}$$
$$M = k\sqrt{L_1 L_2}$$
$$\omega M = \omega k\sqrt{L_1 L_2}$$
$$\begin{aligned}\omega M &= k\sqrt{(\omega L_1)(\omega L_2)} \\ &= 0.5\sqrt{5(20)} \\ &= 5\end{aligned}$$

Thus, the impedance of the mutual inductance is $j\omega M = j5\ \Omega$.

Step 3 of 8

Determine the Thevenin voltage across the terminals *a* and *b*.

Apply the Kirchhoff's voltage law around mesh 1 of the circuit shown in Figure 1.

$$-440\angle 0° - j4\mathbf{I}_1 + j5\mathbf{I}_1 + j\omega M\mathbf{I}_2 + (4+j6)(\mathbf{I}_1 + \mathbf{I}_2) = 0$$

Here, $+j\omega M\mathbf{I}_2$ is considered because the current $\mathbf{I}_2$ flows into the dotted terminal of L_2.

$$-440\angle 0° - j4\mathbf{I}_1 + j5\mathbf{I}_1 + j\omega M(0) + (4+j6)(\mathbf{I}_1 + 0) = 0$$
$$(4+j7)\mathbf{I}_1 = 440\angle 0°$$

Since, current $\mathbf{I}_2$ in the mesh 2 is zero because it is not a closed mesh.

$$\begin{aligned}\mathbf{I}_1 &= \frac{440\angle 0°}{4+j7} \\ &= \frac{440\angle 0°}{8.06226\angle 60.2551°} \\ &= 54.575\angle -60.2551°\ \text{A}\end{aligned}$$

Step 4 of 8

Apply the Kirchhoff's voltage law around mesh 2 of the circuit shown in Figure 1.

$$j2\mathbf{I}_2 + j20\mathbf{I}_2 + j\omega M\mathbf{I}_1 + (4+j6)(\mathbf{I}_2 + \mathbf{I}_1) - \mathbf{V}_{\text{Th}} = 0$$
$$j2(0) + j20(0) + j5\mathbf{I}_1 + (4+j6)(0 + \mathbf{I}_1) - \mathbf{V}_{\text{Th}} = 0$$

$$\begin{aligned}\mathbf{V}_{\text{Th}} &= (4+j11)\mathbf{I}_1 \\ &= (4+j11)(54.575\angle -60.2551°) \\ &= 638.787\angle 9.762°\ \text{V}\end{aligned}$$

Therefore, the Thevenin voltage to the left of the load Z in the circuit is

$\boxed{638.787\angle 9.762°\ \text{V}}$.

Step 5 of 8

Set the voltage source of $440\angle 0^\circ$ to zero. Connect a voltage source of $1\angle 0^\circ$ V across the terminals *a* and *b* and assume that the voltage source delivers current $\mathbf{I}_o$ to the circuit.

Redraw the circuit as shown in Figure 2.

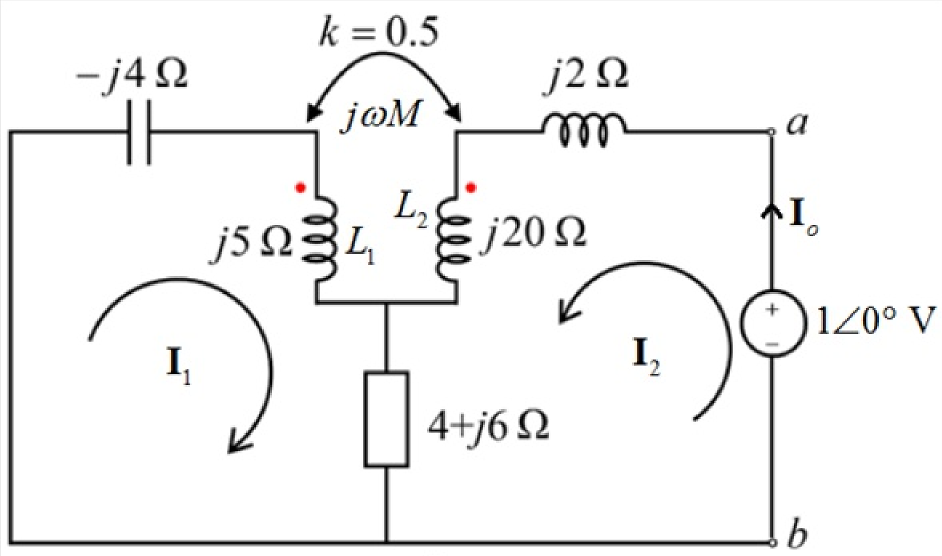


Figure 2

Step 6 of 8

Apply the Kirchhoff's voltage law around mesh 1 of the circuit shown in Figure 2.

$$-j4\mathbf{I}_1 + j5\mathbf{I}_1 + j\omega M\mathbf{I}_2 + (4+j6)(\mathbf{I}_1+\mathbf{I}_2) = 0$$
$$-j4\mathbf{I}_1 + j5\mathbf{I}_1 + j5\mathbf{I}_2 + (4+j6)(\mathbf{I}_1+\mathbf{I}_2) = 0$$

$(4+j7)\mathbf{I}_1 + (4+j11)\mathbf{I}_2 = 0$ ...... (1)

Apply the Kirchhoff's voltage law around mesh 2 of the circuit shown in Figure 2.

$$j2\mathbf{I}_2 + j20\mathbf{I}_2 + j\omega M\mathbf{I}_1 + (4+j6)(\mathbf{I}_2+\mathbf{I}_1) - 1\angle 0^\circ = 0$$
$$j2\mathbf{I}_2 + j20\mathbf{I}_2 + j5\mathbf{I}_1 + (4+j6)(\mathbf{I}_2+\mathbf{I}_1) - 1\angle 0^\circ = 0$$

$(4+j11)\mathbf{I}_1 + (4+j28)\mathbf{I}_2 = 1$ ...... (2)

Observe that the currents $\mathbf{I}_1$ and $\mathbf{I}_2$ are non-zero in the circuit.

Step 7 of 8

Solve equations (1) and (2). Determine current $\mathbf{I}_2$.

Apply the Crammer's rule.

$$\mathbf{I}_2 = \frac{\begin{vmatrix} 4+j7 & 0 \\ 4+j11 & 1 \end{vmatrix}}{\begin{vmatrix} 4+j7 & 4+j11 \\ 4+j11 & 4+j28 \end{vmatrix}}$$
$$= \frac{4+j7}{(4+j7)(4+j28)-(4+j11)^2}$$
$$= \frac{4+j7}{-180+j140-(-105+j88)}$$
$$= \frac{4+j7}{-75+j52}$$
$$= \frac{8.0622\angle 60.2552°}{91.2633\angle 145.265°}$$
$$= 0.0883\angle -85.01° \text{ A}$$
$$= 0.0077 - j0.088 \text{ A}$$

Step 8 of 8

Determine the Thevenin impedance.

$$\mathbf{Z}_{\text{Th}} = \frac{1\angle 0° \text{ V}}{\mathbf{I}_o}$$
$$= \frac{1\angle 0°}{\mathbf{I}_2}$$
$$= \frac{1\angle 0°}{0.0883\angle -85.01°}$$
$$= 11.32\angle 85.01°$$
$$= 0.984 + j11.277\ \Omega$$

Therefore, the Thevenin impedance is $\boxed{0.984 + j11.277\ \Omega}$.